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|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------|-----------------------|----------------|
| Programme      | B. Sc. Computer Science                                                                                                                                                                                                                   |                     |                      |                       |                |
| Course Title   | Advanced Data Structures and algorithms                                                                                                                                                                                                   |                     |                      |                       |                |
| Type of Course | Major                                                                                                                                                                                                                                     |                     |                      |                       |                |
| Semester       | VII                                                                                                                                                                                                                                       |                     |                      |                       |                |
| Academic Level | 400-499                                                                                                                                                                                                                                   |                     |                      |                       |                |
| Course Details | Credit                                                                                                                                                                                                                                    | Lecture<br>per week | Tutorial<br>per week | Practical<br>per week | Total<br>Hours |
|                | 4                                                                                                                                                                                                                                         | 3                   | -                    | 2                     | 75             |
| Pre-requisites | 1. Fundamental Mathematics Concepts: Sets, matrices<br>2. Awareness of Data structures and operations like array, stack, queue<br>3. Fundamentals of Java, C Programming                                                                  |                     |                      |                       |                |
| Course Summary | This course provides an introduction to the ideas, techniques, and applications of advanced data structures) is given in this course. The advanced data structures and its variants like tree, graph, heaps are covered in this syllabus. |                     |                      |                       |                |

#### Course Outcomes (CO):

| CO  | CO Statement                                                                                            | Cognitive Level* | Knowledge Category# | Evaluation Tools used                                  |
|-----|---------------------------------------------------------------------------------------------------------|------------------|---------------------|--------------------------------------------------------|
| CO1 | Understand the concepts of advanced data structures like tree, graphs, heaps.                           | U                | C                   | Instructor-created exams / Quiz                        |
| CO2 | Understand familiarity with algorithmic techniques such as brute force, greedy, and divide and conquer. | U                | C                   | Practical Assignment / Observation of Practical Skills |
| CO3 | Understand Asymptotic analysis (big-O notation, time and space complexity).                             | U                | F                   | Seminar Presentation / Group Tutorial Work/ Viva Voce  |
| CO4 | Application of advanced abstract data type (ADT) and data structures in solving real world problems.    | AP               | P                   | Instructor-created exams / Home Assignments            |

|                                                                                                                                                                                                |                                                                                                                                           |    |   |                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----|---|--------------------------------------------------------------|
| CO5                                                                                                                                                                                            | Effectively combine fundamental data structures and algorithmic techniques in building a complete algorithmic solution to a given problem | Ap | P | Writing assignments/<br>Instructor-created exams/ practicals |
| CO6                                                                                                                                                                                            | Apply Concepts of data structures in real world problem solving                                                                           | Ap | P | Case Study/ mini Project/ practicals                         |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M) |                                                                                                                                           |    |   |                                                              |

### Detailed Syllabus:

| Module    | Unit                                                                           | Content                                                                                                                                            | Hrs<br>(45+30) | Marks<br>(70) |
|-----------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------|
| <b>I</b>  | <b>Introduction to data structures and analysis of Quality of an algorithm</b> |                                                                                                                                                    | <b>9</b>       | <b>12</b>     |
|           | 1                                                                              | Data structure - definition - types & operations, characteristics of data structures                                                               | 2              |               |
|           | 2                                                                              | Abstract Data Type (ADT) – algorithms - concepts - definition - objectives of algorithms -                                                         | 1              |               |
|           | 3                                                                              | Quality of an algorithm - space complexity and time complexity of an algorithm.                                                                    | 2              |               |
|           | 4                                                                              | Growth of Functions: Asymptotic notations, Cost estimation based on key operations- Big Oh, Big Omega, Little Oh, Little Omega and Theta notations | 3              |               |
|           | 5                                                                              | Algorithm Design: Introduction, Steps in developing algorithm, Methods of specifying an algorithm                                                  | 1              |               |
| <b>II</b> | <b>Basic Technique for Design of Efficient Algorithm</b>                       |                                                                                                                                                    | <b>11</b>      | <b>18</b>     |
|           | 6                                                                              | Brute Force approach (String pattern matching)                                                                                                     | 1              |               |
|           | 7                                                                              | Divide-and-Conquer approach (Merge sort)                                                                                                           | 1              |               |
|           | 8                                                                              | Branch-and-Bound technique (Knapsack problem)                                                                                                      | 2              |               |
|           | 9                                                                              | Greedy approach (Kruskal's algorithm and Prim's Algorithm)                                                                                         | 3              |               |

|            |                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |           |
|------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
|            | 10                                                                                                       | Dynamic Programming (Longest Common Subsequence)                                                                                                                                                                                                                                                                                                                                                                                                                                | 2         |           |
|            | 11                                                                                                       | Backtracking (Sum of subsets problem)                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2         |           |
| <b>III</b> | <b>Linked lists - operations and implementations</b>                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>12</b> | <b>20</b> |
|            | 12                                                                                                       | Introduction to Singly Linked list and its operations                                                                                                                                                                                                                                                                                                                                                                                                                           | 2         |           |
|            | 13                                                                                                       | Circular Linked list and its operations                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3         |           |
|            | 14                                                                                                       | Doubly Linked list and its operations                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2         |           |
|            | 15                                                                                                       | Circular Doubly Linked list and its operations                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2         |           |
|            | 16                                                                                                       | Recursive lists, heterogeneous lists, deterministic skip lists- Creation & Searching                                                                                                                                                                                                                                                                                                                                                                                            | 3         |           |
| <b>IV</b>  |                                                                                                          | <b>Non-linear Data Structures</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>13</b> | <b>20</b> |
|            | 17                                                                                                       | Binary search trees - traversals and operations on BST                                                                                                                                                                                                                                                                                                                                                                                                                          | 3         |           |
|            | 18                                                                                                       | AVL tree, Red Black Tree (concept only)                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2         |           |
|            | 19                                                                                                       | Balanced trees - M-way trees - B Tree (Concepts only)                                                                                                                                                                                                                                                                                                                                                                                                                           | 1         |           |
|            | 20                                                                                                       | Graphs - representation of graphs                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1         |           |
|            | 21                                                                                                       | Graphs- operations - traversals and their implementation.                                                                                                                                                                                                                                                                                                                                                                                                                       | 2         |           |
|            | 22                                                                                                       | Heap structures- Min-Max heaps - Deaps - leftist heaps - binomial heaps (concepts only) - applications                                                                                                                                                                                                                                                                                                                                                                          | 3         |           |
| <b>V</b>   | <b>Practical Implementations of Data structures and its operations in Java or C programming Language</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>30</b> |           |
|            | 1                                                                                                        | <ul style="list-style-type: none"> <li>• Implementation of linear linked list</li> <li>• Implementation of circular linked list</li> <li>• Implementation of doubly linked list</li> <li>• Implementation of BST operations</li> <li>• Implementation of Depth First Search using graph</li> <li>• Implementation of Breadth First Search using graph</li> <li>• Implementation of max heap and delete a node from it.</li> <li>• Sort a set of data using Heap tree</li> </ul> | 25        |           |
|            | 2                                                                                                        | Case Study/ Project                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5         |           |

## References

1. Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, Data Structures and Algorithms, Addison-Wesley, ISBN: 978-0201000238.

2. Horowitz E and Sahni S, Fundamentals of Data Structures, Computer Science Press, ISBN: 9780716780427.
3. Ellis Horowitz, Sartaj Sahni and Susan Anderson-Freed, Fundamentals of Data Structures in C, Silicon Press, ISBN: 0929306406.
4. Thomas H Cormen, Charles E Leiserson, and Ronald L Rivest, Introduction to Algorithms, 3rd Edition, Prentice Hall of India Private Limited, New Delhi, ISBN: 9780262033848
5. Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, The Design and Analysis of Computer Algorithms, 1st Edition. Addison Wesley, ISBN: 0534915728

#### Mapping of COs with PSOs and POs :

|      | PSO1 | PSO2 | PSO3 | PSO4 | PSO5 | PSO6 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 |
|------|------|------|------|------|------|------|-----|-----|-----|-----|-----|-----|
| CO 1 | 2    | 1    | 3    | 3    | 2    | 1    |     |     |     |     |     |     |
| CO 2 | 2    | 1    | 3    | 3    | 2    | 1    |     |     |     |     |     |     |
| CO 3 | 2    | 1    | 3    | 3    | 2    | 1    |     |     |     |     |     |     |
| CO 4 | 3    | 1    | 3    | 3    | -    | -    |     |     |     |     |     |     |
| CO 5 | 1    | 1    | 3    | 3    | 3    | 1    |     |     |     |     |     |     |
| CO 6 | 2    | 1    | 3    | 3    | 3    | 1    |     |     |     |     |     |     |

#### Correlation Levels:

| Level | Correlation        |
|-------|--------------------|
| -     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

#### Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

#### Mapping of COs to Assessment Rubrics :

|  | Internal Exam | Assignment | Practical Evaluation | End Semester Examinations |
|--|---------------|------------|----------------------|---------------------------|
|--|---------------|------------|----------------------|---------------------------|

|      |   |   |   |   |
|------|---|---|---|---|
| CO 1 | ✓ | ✓ | ✓ | ✓ |
| CO 2 | ✓ | ✓ | ✓ | ✓ |
| CO 3 | ✓ | ✓ | ✓ | ✓ |
| CO 4 | ✓ | ✓ | ✓ | ✓ |
| CO 5 | ✓ | ✓ | ✓ | ✓ |
| CO 6 | ✓ | ✓ | ✓ |   |